

Department of Computer Science and Engineering
Jashore University of Science and Technology
Semester Final Examination-2023
B.Sc. (Engg.) 2nd Year 2nd Semester, Session: 2020-21
Course Code: CSE 2203
Course Title: Database Management System
Time: 3.00 hours
Marks: 72

[N.B. Answer any 6 set of questions from the following 8 set of questions. The figures in the right margin indicate full marks.]

1.

- (a) Define Database Management System (DBMS). Mention the advantages of DBMS over the conventional file processing system. **(3 Marks)**
- (b) Write down any two major responsibilities of a database administrator. **(2 Marks)**
- (c) "All candidate keys are super keys, but all super keys are not candidate keys" - Justify this statement. **(3 Marks)**
- (d) Explain the term Distributed Database Management System (DDBMS). Mention the issues to be considered in the design of DDBMS. **(4 Marks)**

2.

- (a) From the following, identify DDL, DML, Logical design, and Physical design SQLs: **(6 Marks)**
 - 1. CREATE INDEX idx_lastname ON Persons (LastName);
 - 2. DROP INDEX index_name ON table_name;
 - 3. DROP TABLE table_name;
 - 4. SELECT CustomerName, ContactName INTO CustomersBackup2017 FROM Customers;
- (b) "A relational data model relates the data instances" - Justify. **(2 Marks)**
- (c) Explain query processor. **(4 Marks)**

3.

- (a) Consider the following relational schema of an employee database: **(6×2=12 Marks)**
 - Employee (**employee_name, street, city**)
 - Works (**employee_name, company_name, salary**)
 - Company (**company_name, city**)
 - Manages (**employee_name, manager_name**)

Write Structured Query Language (SQL) to:

- 1. Find the names of all employees who work for First Bank Corporation.
- 2. Find all employees in the database who live in the same cities as the companies for which they work.

3. Find all employees in the database who live in the same cities and on the same streets as their managers.
4. Find all employees who earn more than the average salary of all employees of their company.
5. Find the company that has the smallest payroll.
6. Display all managers of First Bank Corporation with a 10 percent raise in salary.

4.

(a) Consider the following schema: **(8 Marks)**

- Student (**RollNo, FirstName, EnglishMarks, PhysicsMarks, ChemistryMarks, MathsMarks, TotalMarks**)

You are required to:

- Create a trigger for each row to calculate and set the value of TotalMarks while inserting values in other columns. (TotalMarks = EnglishMarks + PhysicsMarks + ChemistryMarks + MathsMarks)
- (b) Define functional dependency. Describe different types of functional dependencies with proper examples. **(4 Marks)**

5.

Design a database for an airline. The database must keep track of customers and their reservations, flights and their status, seat assignments on individual flights, and the schedule and routing of future flights. **(4+4+4=12 Marks)**

Your design should include:

1. An E-R diagram.
2. A set of relational schemas (derived from E-R diagram).
3. A list of constraints, including primary-key and foreign-key constraints.

6.

(a) Normalize the following schema with given constraints to 4NF: **(8 Marks)**

- books (**accession_no, isbn, title, author, publisher**)
- users (**user_id, name, dept_id, dept_name**)
- accession_no → isbn
- isbn → title
- isbn → publisher
- isbn → author
- user_id → name
- user_id → dept_id
- dept_id → dept_name

(b) Explain ACID properties. **(4 Marks)**

7.

(a) "Consistency of the database must be preserved even with the expense of concurrency" - Justify with the following schedule: **(6 Marks)**

- **T1:** Read(A), $A = A - 50$, Write(A), Read(B), $B = B + 50$, Write(B), Commit
- **T2:** Read(A), Read(B), $Z = A + B$, Write(Z), Commit

(b) "Cascading rollback may waste a massive amount of resources" - Explain with respect to the following schedule: **(6 Marks)**

- **T1:** Read(A), Read(B), Write(A)
- **T2:** Read(A), Read(B)
- **T3:** Read(A)

8.

(a) Explain the evolution of the locking protocol based on the following schedule: **(6 Marks)**

T1	T2	Concurrency Control Manager
Lock-X(B)		Grant-X(B, T1)
Read(B)		
$B := B - 50$		
Write(B)		
Unlock(B)		
	Lock-S(A)	Grant-S(A, T2)
	Read(A)	
	Unlock(A)	
	Lock-S(B)	Grant-S(B, T2)
	Read(B)	
	Unlock(B)	
	Display(A+B)	
Lock-X(A)		Grant-X(A, T1)
Read(A)		
$A := A + 50$		
Write(A)		
Unlock(A)		

(b) "Locking protocols may produce deadlock. But deadlock is better than inconsistency" - Justify with the following transactions P1, P2, P3, P4, and P5. **(6 Marks)**
